

SARA-Net: Multimodal SAR and Climate Data Fusion for Flood Risk Assessment in the Sundarbans

Abstract—The Sundarbans is the biggest mangrove forest in the world, which is important for protecting the coastal areas of Bangladesh from cyclones, storm surge and floods. But with the effects of climate change, sea-level rise and alteration in precipitation, this ecosystem is becoming more vulnerable to flooding. Nevertheless, the existing flood monitoring systems are mostly based on optical satellite imagery, which is not suitable during monsoon season due to high cloud cover. Furthermore, previous research has not been able to merge and integrate satellite radar images and climate-related information into a consistent deep learning framework for this area. In response to these issues, a SARA-Net (Sundarbans Adaptive Risk Assessment Network) framework is proposed in this study which consists of two streams: one deep learning stream using SAR imagery acquired by Sentinel-1 and another deep learning stream using rainfall and river-level data from three districts between 2014 and 2025. The proposed model reached an accuracy of 76.98% with an ROC-AUC of 0.771 on the fixed group-disjoint test set. The Visual-Only model achieved 76.34%, while the Climate-Only model achieved 48.08%. Nested grouped cross-validation yielded an accuracy of $74.80 \pm 12.31\%$, indicating substantial group-level variability. Among the study areas, Satkhira is found to be the most vulnerable, with a risk score of 0.88. The benchmark uses a synthetic mechanism-informed flood-impact target. To improve model interpretability, Explainable AI (XAI) techniques based on LayerCAM are employed. The resulting heatmaps show that the model primarily focuses on riverbanks and damaged embankments when identifying flood-prone areas. These findings demonstrate the potential of SARA-Net as an interpretable, all-weather framework for supporting flood risk assessment and disaster management in vulnerable coastal regions.

Index Terms—Flood Risk Assessment, Synthetic Aperture Radar, Multimodal Deep Learning, Explainable AI, Mangrove Ecosystem

I. INTRODUCTION

The Sundarbans, shared by Bangladesh and India, is the largest mangrove forest in the world, covering an area of approximately 10,000 square kilometers of the Ganges–Brahmaputra delta [1]. This unique forest plays a vital role in protecting coastal communities from cyclones that frequently form in the Bay of Bengal. It also supports local livelihoods through fishing, honey collection, and timber harvesting. Its significance is growing yet the Sundarbans are under increasing environmental stress. Climate change is affecting the balance between freshwater and saltwater systems, leading to sea-level rise, irregular rainfall patterns, and higher soil salinity. These changes are damaging both the forest ecosystem and the nearby agricultural land [2], [1].

The effects of environmental degradation are becoming more visible each year. Studies using Landsat and Sentinel-2 satellite images have shown a steady decline in dense forest cover over the last 50 years [5], [6]. Survey results indicate that more than 90% of coastal residents face climate-related risks, while many have limited resources and knowledge to cope with these challenges [11]. Monitoring these environmental changes remains challenging. During the monsoon season, thick cloud cover often blocks optical satellite observations for long periods [7]. Consequently, important information is often unavailable when flood monitoring is most critical. Furthermore, many previous studies have focused on individual factors such as salinity [9] or forest loss [8], rather than developing a complete all-weather flood prediction system.

These challenges highlight an important research gap. Existing studies have not effectively combined all-weather Synthetic Aperture Radar (SAR) imagery with climate records in a single deep learning framework for flood risk prediction in the Sundarbans. The SAR technology detects microwave signals that can penetrate cloud cover and can provide reliable observations during the monsoon season [30]. Deep learning techniques have been used for flood detection in Bangladesh, but many models have been making predictions without any explanation for the results. Many flood-related factors are not known, including damaged embankments or blocked drainage channels [21], [22]. The opacity hinders the utility of AI for disaster management and emergency planning. Explainable AI (XAI) can be applied to create heatmaps that can identify flood-related features such as embankment damage and saturated soils, improving the transparency of the model. These visual explanations provide a way to differentiate infrastructure-related flooding from natural waterlogging [29].

These limitations are overcome by this study, which introduces SARA-Net. This is the dual-stream (Sundarbans Adaptive Risk Assessment Network). This model combines Sentinel-1 SAR imagery with rainfall and river-level data collected from Satkhira, Khulna, and Bagerhat between 2014 and 2025. The benchmark uses a synthetic mechanism-informed flood-impact target. All-weather, multi-source monitoring is required for such conditions. By providing reliable flood risk assessments for different locations, SARA-Net can help decision-makers strengthen infrastructure, improve disaster preparedness, and support long-term climate adaptation across the region. To address the identified research gaps, this study makes four key contributions:

- 1) Proposing SARA-Net, a dual-stream system that enhances flood risk assessment in the Sundarbans by combining climate data and Sentinel-1 SAR imagery.
- 2) Developing an all-weather flood monitoring system that works even during the monsoon season, when clouds frequently obscure optical satellite imagery.
- 3) Detecting flood risks by identifying silent flood episodes and hydrological decoupling.
- 4) Using LayerCAM to improve model interpretability by visualizing the spatial features that influence flood-risk predictions.

The remainder of this paper is structured as follows. A review of existing studies on Sundarbans monitoring, SAR-based flood detection, and explainable deep learning is presented in Section II. Section III introduces the dataset, outlines the preprocessing steps, and describes the proposed SARA-Net framework. The experimental results and key observations are discussed in Section IV. Finally, Section V concludes the study and highlights opportunities for future research.

II. RELATED WORK

Recent publications related to the Sundarbans have emphasised the importance of improved environmental monitoring. Islam [11] did coastal village studies to understand the human component of climate change: 79% of the people were not able to cope with it, and 90% were exposed to climate change risks. Likewise, Chakma et al. [12] interviewed the local people in Satkhira District and determined that around 30% of the population had to shift their livelihoods, usually from their traditional work, due to saltwater intrusion. These studies are useful background information, but they are carried out on smaller areas and are not able to explain what is happening throughout the forest ecosystem.

Satellite images, however, have superseded observation of the physical changes. Enan et al. [14] showed, using Landsat images, that cyclones are the single most common cause of forest loss. Roy et al. [6] and Hossain et al. [5] created forest loss maps for the past decades based on Sentinel-2 data. Wahid et al. [2] and Nandi et al. [15] addressed water quality issues, both of which have found that salinity increases are in an acute phase in the central Sundarbans. However, optical satellite-based studies remain limited during monsoon periods because persistent cloud cover reduces image availability. Some authors tried to combine optical and radar data but merging optical and radar datasets is a challenging technical problem [17] [8].

However, these challenges have been overcome by the power of deep learning for analysing environmental data. In a previous work, AI techniques such as CNN models have been proved to be effective in fusing multiple modalities of data [10]. On this basis, Giezendanner et al. [31] built a CNN-LSTM-based model, incorporating both Sentinel-1 and MODIS (Moderate Resolution Imaging Spectroradiometer) data to reconstruct the flood inundation in Bangladesh for 20 years, showing the significant value of temporal context to be added to the spatial value of the model. Traditional statistical

techniques still outperform some hybrid techniques in flood monitoring, according to Rahman et al. [18]. Hemel and Sakib [24] evaluated the performance of the models Gradient Boosting, Random Forest, XGBoost, and LSTM for flood prediction in Bangladesh, where the researchers discovered that the Gradient Boosting model has an accuracy of 95.38%, and further used SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) to make the model more interpretable. Though significant progress in AI, the "black box" issue is still present. To address this black-box problem, Pradhan et al. [29] integrated XAI (SHAP) into CNN-based flood susceptibility models and found that elevation and distance to streams are the most important flood predictors. To solve this problem, Sanderson et al. [28] proposed an explainable fusion model of SAR and optical images where transparency and accuracy are achieved simultaneously.

These studies point out two key gaps in existing research. Most earlier work depends on optical satellite images, which often do not work well during cloudy or rainy weather. In contrast, SAR radar data is used here to support more reliable flood monitoring in all weather conditions [7]. Another gap is the limited transparency in AI-based flood prediction models [18]. To improve this, LayerCAM visualization is used to show which geographic features, such as damaged embankments and waterlogged soil, are influencing the model's decisions [23]. This study further evaluates multimodal fusion under group-disjoint testing and controlled baseline comparisons.

III. METHODOLOGY

This study presents a Multimodal Data Fusion framework to predict flood risk in the Sundarbans by integrating Sentinel-1 SAR satellite imagery with environmental climate records. The resulting dataset contains 4,650 samples across 36 district-year groups.

1) *Visual Data (SAR)*: Sentinel-1 SAR imagery from the SAR image of Sunderbans dataset [25] is used for its all-weather cloud-penetrating capability, which makes it uniquely suited for flood monitoring during the monsoon season when optical satellites are rendered blind by cloud cover [30]. The dataset provides three distinct processing levels: (i) *Enhanced Visualization* as RGB composites for qualitative analysis of significant weather events such as cyclones; (ii) *SAR Urban* mode, designed for monitoring infrastructure, embankments, and human encroachment; and (iii) *VH Polarization*, which is sensitive to vegetation structure and supports the monitoring of mangrove health and biomass.

2) *Climate*: Following the approach used in the Bangladesh Climate Change Simulation Dataset [26], synthetic climate records are created for Satkhira, Khulna, and Bagerhat using real-world events, including the peak rainfall of 4537 mm in 2018 and the silent flood episode in 2022. Historical records spanning 2014 to 2025 are analyzed to estimate flood impact scores and understand changes in regional flood patterns.

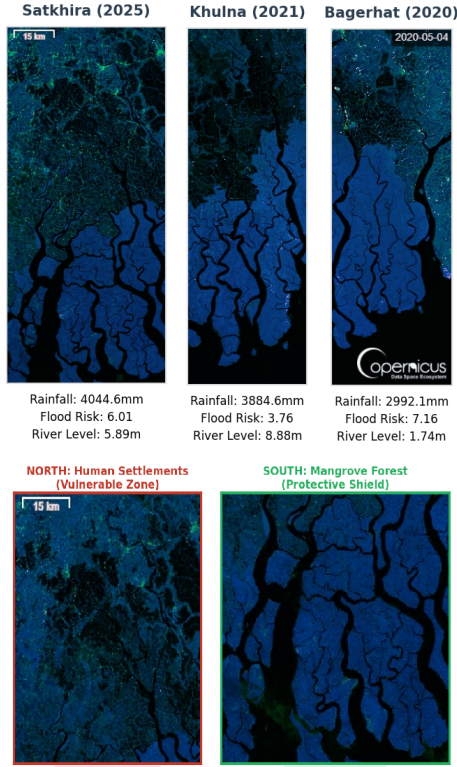


Fig. 1. Preprocessed SAR samples.

A. Data Preprocessing and Labeling

A systematic preprocessing pipeline is developed to synchronize the visual and climate data streams before model training. First, EasyOCR (Optical Character Recognition) is applied to extract year stamps directly from the raw satellite image frames, enabling accurate temporal alignment between SAR images and their corresponding climate records for the same year and district. Second, a spatial splitting algorithm is applied to each image to separate ecologically distinct zones. The top 55% of each cropped image is extracted as the *North Zone*, representing human settlement areas, and the remaining 45% is extracted as the *South Zone*, representing the mangrove forest shield. Third, each image is vertically cropped into three district regions based on fixed proportional coordinates: Satkhira (0–40%), Khulna (40–70%), and Bagerhat (70–100%) of the image width, corresponding to the Western, Central, and Eastern ecological ranges, respectively. Finally, the three SAR sensor modes are processed separately and merged with their matching climate records to form complete multimodal samples. Each sample is labeled as *High Risk* (1) if its flood impact score exceeded the dataset median, and *Safe* (0) otherwise. The target is synthetic and should not be interpreted as measured flood impact.

B. Data Augmentation

Various data augmentation techniques are applied during training to improve generalization and prevent overfitting. Random horizontal flips and affine translations are used to

simulate natural variations in the positioning of the satellite images. A feature noise injection approach is also used during the training phase, by randomly substituting 10% of the climate features in the climate feature vector with Gaussian noise ($\sigma = 0.05$), mimicking real-world sensor measurement noise. Only resizing and normalization transforms are used in validation and test sets to assure unbiased evaluation.

C. Proposed Architecture (SARA-Net)

A dual-stream neural network called SARA-Net is developed to handle the multimodal input. The model consists of three primary components, as illustrated in Figure 2.

1) *Visual Stream (ResNet-18)*: The model is based on a pretrained ResNet-18 backbone that is used to process 224×224 satellite images. The last classification layer is replaced by an identity mapping to obtain a feature vector with 512 dimensions. To reduce overfitting, a Dropout layer ($p = 0.5$) is used after the backbone. ResNet-18 is used as the visual backbone because it performs well on regional image datasets with limited training data and often outperforms custom CNN models trained from scratch [32].

2) *Context Stream (Multi-Layer Perceptron)*: A Multi-Layer Perceptron (MLP) processes the 5-dimensional environmental input vector, which consists of normalized rainfall, normalized river water level, and a one-hot encoded sensor type indicator representing the three SAR processing modes (Enhanced Visualization, SAR Urban, VH Polarization). The MLP consists of two fully connected layers ($5 \rightarrow 64 \rightarrow 32$) with Batch Normalization and ReLU activation, producing a 32-dimensional context embedding.

3) *Fusion and Classification Head*: The 512-dimensional visual vector and 32-dimensional context vector are concatenated into a single 544-dimensional representation. This combined vector is passed through a series of dense layers ($544 \rightarrow 128 \rightarrow 64 \rightarrow 1$) with ReLU activation and Dropout ($p = 0.3$). The output layer produces a continuous probability score between 0 (Safe) and 1 (High Risk). The fusion results indicate a SAR-dominant fusion behavior rather than a separate rule-based override mechanism. This suggests that the model relies more strongly on SAR-derived visual features while using climate variables as supporting contextual information.

D. Explainable AI Implementation

Considering the black box issue in deep learning, an XAI module is incorporated into the model based on the LayerCAM technique [29] [23]. Forward and backward hooks are attached to the last convolutional layer of the ResNet-18 backbone to collect activation maps and gradient features during inference. The outputs are then aggregated to generate spatial heatmaps indicating the geographic areas that most influence each flood risk prediction, such as riverbanks, broken embankments, and settlement edges.

E. Experimental Setup

SARA-Net is implemented using PyTorch and trained on an NVIDIA T4 GPU via Google Colab. The dataset is

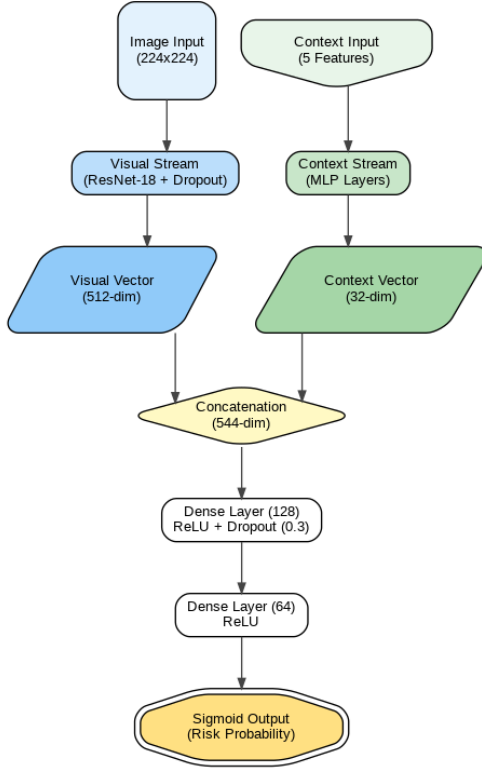


Fig. 2. SARA-Net architecture.

divided into group-disjoint training, validation, and testing sets, with 24, 6 and 6 district-year groups, respectively. The model is trained for 10 epochs with a batch size of 32, using Binary Cross-Entropy (BCE) loss. Adam optimizer with learning rate 5×10^{-5} , and weight decay 1×10^{-4} (regularization). The learning rate is reduced by 0.5 when validation accuracy has not improved for 4 epochs using a ReduceLROnPlateau scheduler, such that the learning rate is decreased slowly as necessary while maintaining stable validation convergence. Group-disjoint splitting prevents samples from the same district-year group from appearing across evaluation sets. Nested grouped cross-validation is additionally used to assess group-level generalization.

IV. RESULTS AND DISCUSSION

The performance of SARA-Net is evaluated using the fixed group-disjoint test set, where it achieves an accuracy of 76.98% and an ROC-AUC score of 0.771. These results suggest that the model is effective in identifying flood risk patterns in the Sundarbans. The following sections discuss the model's performance, the contribution of each data source, key findings, regional vulnerability patterns, and the explainability analysis.

A. Baseline Comparison

The performance of SARA-Net is compared with two single-modality models: a Climate-Only MLP and a Visual-Only ResNet-18. In the Climate-Only model, the accuracy

TABLE I
BASELINE PERFORMANCE COMPARISON

Model	Modality	Acc (%)	Precision	Recall	F1
Climate-Only (MLP)	Climate	48.08	0.45	0.35	0.39
Visual-Only (ResNet-18)	SAR Images	76.34	0.71	0.88	0.78
SARA-Net (Proposed)	Fusion (Both)	76.98	0.68	0.97	0.80

TABLE II
BACKBONE COMPARISON

Backbone	Acc (%)	AUC
ResNet-18	76.98	0.771
ResNet-50	78.26	0.786
DenseNet-121	76.21	0.770
EfficientNet-B0	71.99	0.778

was 48.08%, as presented in Table I, which indicates that the climate information is not enough to fully capture the flood risk in the study area. The Visual-Only model gave an accuracy of 76.34%, indicating a good prediction potential of SAR imagery. SARA-Net has the highest accuracy (76.98%). This improvement is consistent with the results of Li et al. [10] showing that heterogeneous data type fusion consistently enhances the performance of deep learning in environmental monitoring-related applications. The confusion matrix validates the classification performance of the proposed model.

SARA-Net achieves a slightly higher accuracy than the Visual-Only model, while both substantially outperform the Climate-Only model.

B. Backbone Comparison

Different pretrained visual backbones are evaluated to assess the effect of the visual feature extractor. ResNet-18 achieves 76.98% accuracy and 0.771 AUC, while ResNet-50 achieves the highest accuracy of 78.26% and an AUC of 0.786. DenseNet-121 and EfficientNet-B0 achieve accuracies of 76.21% and 71.99%, respectively.

ResNet-50 achieved the highest accuracy, while ResNet-18 provided competitive performance with a lighter backbone.

C. Grouped Cross-Validation

Nested grouped cross-validation is used to assess group-level generalization. SARA-Net achieves an accuracy of $74.80 \pm 12.31\%$ and an AUC of 0.688 ± 0.211 , indicating substantial group-level variability. The Visual-Only model achieves $70.28 \pm 15.29\%$ accuracy and 0.713 ± 0.184 AUC.

D. Hydrological Decoupling Analysis

An important result is the understanding of hydrological decoupling, which is the lack of a direct relationship between flood risk and intensity of rainfall. The correlation between the annual rainfall and the flood risk based on the prediction of the flood risk from 2014 to 2025 is shown in Fig. 3. A low

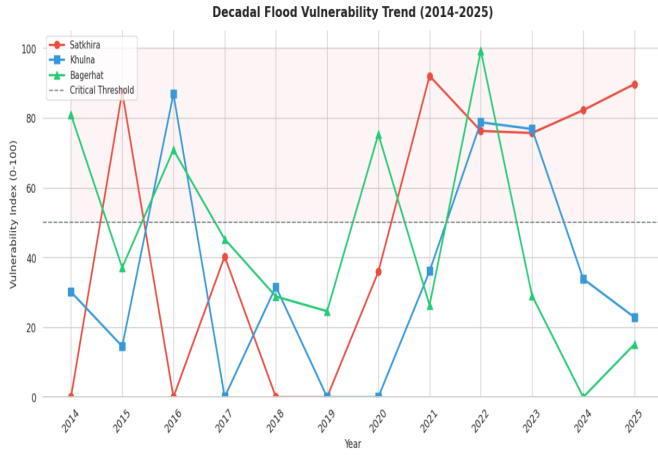


Fig. 4. Regional flood risk.

flood risk score is predicted for Satkhira despite the highest annual rainfall of the study period (4537 mm) in 2018. A high flood risk score was, however, forecast in 2022 with relatively low rainfall. This event is interpreted as a possible silent flood pattern, rather than a confirmed real-world flood event, where high flood risk may be associated with drainage failure and embankment weakness rather than rainfall intensity alone [24].

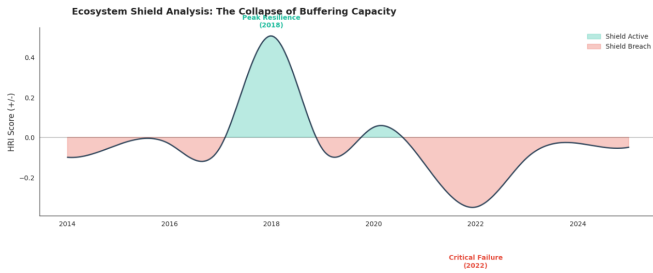


Fig. 3. Flood risk and rainfall trends.

E. Regional Vulnerability and Ecosystem Shield Analysis

The Hydrological Resilience Index (HRI) is used to determine the flood buffering capacity of the Sundarbans ecosystem. The positive values of the HRI in the mangrove forest between 2014 and 2020 demonstrated that the mangrove forest was an effective natural barrier against flooding. There is a marked decrease in HRI after 2020, especially in the vicinity of the 2022 silent flood event. This trend indicates a progressive loss of the ecosystem's protective ability. Similar environmental degradation has been reported in recent remote sensing studies [5], [27]. The regional assessment identified Satkhira as the most vulnerable district with a flood risk score of 0.88. Lower risk scores of 0.65 and 0.42 are observed for Khulna and Bagerhat (Figure 4), respectively. These results are consistent with previous studies that reported greater environmental stress and infrastructure vulnerability in the western Sundarbans [2], [9].

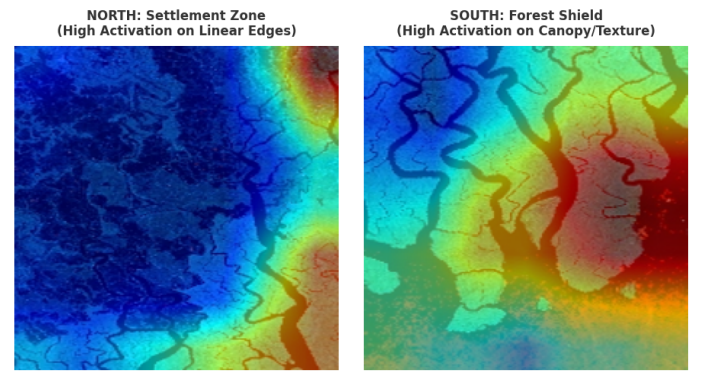


Fig. 5. LayerCAM visualizations.

F. Visual Explainability (LayerCAM)

LayerCAM heatmaps are generated to examine the factors influencing model predictions and to improve model transparency [28], [29]. The heatmaps identify major geographical characteristics, like riverbanks, embankments, the limits of settlements and tidal channels (Figure 5). The use of synthetic aperture radar data, particularly Sentinel-1, also yields strong activation around riverbanks and settlement areas in the North Zone, highlighting locations that pose a higher risk of flooding. The South Zone primarily focuses on mangrove canopy structures, while tidal creek networks are also explored. These results indicate that, in SARA-Net predictions, important environmental features are utilized rather than random image patterns.

V. CONCLUSION

This study addresses the high-stakes issue of accurate flood prediction in one of the world's most climate-vulnerable areas, Sundarbans. Existing methods are limited because these approaches often depend on optical satellite images that fail during the monsoon season due to cloud cover, or on rainfall data alone, which is not sufficient to capture the complex flood behavior in delta regions. Due to these limitations, existing methods cannot accurately classify flood events that are not triggered by precipitation (e.g. impeded drainage and/or breached embankments). Satkhira is the most vulnerable area, with risk score 0.88 as analysed through this report.

To address these problems, SARA-Net is proposed, a dual-stream deep learning model that integrates Sentinel-1 SAR satellite images with climate data to enhance flood vulnerability assessment across the Sundarbans region. It is compared against baseline models, and further backbone comparisons are carried out to demonstrate the significance of each data source. Explainable AI (LayerCAM) is also used to ensure that the model focuses on meaningful physical features such as river networks and settlement boundaries.

The results show that SARA-Net achieves 76.98% accuracy and an AUC of 0.771 on the fixed group-disjoint test set. Nested grouped cross-validation yields an accuracy of $74.80 \pm 12.31\%$, indicating substantial group-level variability. SARA-Net achieves a slightly higher accuracy than the image-only

model, while both substantially outperform the climate-only model. It is also able to identify complex situations such as hydrological decoupling, where flood risk occurs without heavy rainfall. Future improvements include developing a lightweight version for edge devices, enabling real-time prediction using live Sentinel-1 data, and expanding the system to other vulnerable coastal delta regions.

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